

FEDERAL UNIVERSITY OF TECHNOLOGY, OWERRI
SCHOOL OF BIOLOGICAL SCIENCES
DEPARTMENT OF FORENSIC SCIENCE
FRS 309 - Data Analytics and Artificial Intelligence for Biological Sciences (1
Unit)

Biological Data Analysis in Forensic Science

1. Overview of Biological Data Types (The "Omics" Revolution)

In forensic science, we are moving beyond simple DNA profiling to "Molecular Phenotyping." This involves several layers of biological data:

A. Genomics (The Blueprint)

- **Definition:** The study of the complete set of DNA within an organism.
- **Forensic Application:** Beyond CODIS markers (STRs), genomics now includes Single Nucleotide Polymorphisms (SNPs) for Ancestry Informative Markers (AIMs) and Externally Visible Characteristics (EVCs) like eye, hair, and skin color.
- **Data Type:** Sequence data (FASTA/FASTQ), variant files (VCF).

Note:

DNA profiling is a forensic technique used to identify individuals by analyzing specific, highly variable regions of their genetic code to create a unique biological "fingerprint."

Molecular profiling is a method of testing a sample of tissue, blood, or other body fluid to identify specific genes, proteins, or other molecules that provide a detailed "biological map" of a disease, such as cancer.

CODIS markers are a specific set of Short Tandem Repeats (STRs)—highly variable regions of DNA—used by the FBI and forensic labs to create unique genetic profiles for human identification. Because the number of repeats at these specific loci varies significantly between individuals, they provide a powerful statistical tool for linking suspects to crime scene evidence.

- **Data Type:** Sequence data (FASTA/FASTQ), variant files (VCF).

Explaining Genomics Data Types

When scientists look at DNA, they use different "file formats" depending on what they are looking for. Here is how to understand FASTA, FASTQ, and VCF files using everyday analogies.

1. FASTA: The "Master Copy" (The Instruction Manual)

Imagine you have a giant instruction manual for building a human.

- **What it is:** A FASTA file is a simple text file that contains the sequence of DNA letters (A, C, G, and T).
- **The Layman's Terms:** Think of this as the "**Final Draft**" of a book. It's the clean, finished version of the genetic code without any messy notes or corrections. It tells you exactly what the sequence is, from start to finish.

2. FASTQ: The "Rough Notes" (The Scanned Pages)

Imagine you are trying to read that instruction manual, but the pages are torn into millions of tiny pieces, and you're using a camera to take pictures of them.

- **What it is:** A FASTQ file is like a FASTA file, but with one major addition: **Quality Scores**. It tells you the DNA sequence *and* how confident the machine is that it read each letter correctly.
- **The Layman's Terms:** Think of this as a "**Photo of a handwritten note.**" It shows you the text, but it also shows you if the ink is smudged or if the handwriting is hard to read. It helps scientists know which parts of the data are reliable and which might be mistakes.

3. VCF (Variant Call Format): The "List of Differences" (The Red Pen)

Imagine you are comparing your instruction manual to a standard "Reference" manual that everyone is supposed to have.

- **What it is:** A VCF file doesn't show the whole DNA sequence. Instead, it only lists the places where *your* DNA is different from the standard reference.
- **The Layman's Terms:** Think of this as a "**List of Typos.**" Instead of printing the whole book again, the VCF file just says: *"On page 40, the third word should be 'Blue' instead of 'Red'."* It's a shorthand way of showing what makes an individual unique, such as eye color, hair texture, or a risk for a certain disease.

Summary Table

File Type	Analogy	What it tells you
FASTA	The Clean Book	The actual letters of the genetic code.
FASTQ	The Smudged Photo	The letters + how sure we are that we saw them right.
VCF	The List of Typos	Only the spots where you are different from the average person.

B. Proteomics (The Functional Players)

- Definition: The large-scale study of proteins, which are the functional products of genes.
- Forensic Application: Proteins are more stable than DNA in certain environments (e.g., hair shafts, ancient bones). Proteomic profiling can identify the species of origin or even identify a suspect based on genetically variant peptides in hair.
- Data Type: Mass spectrometry (MS) spectra, protein sequence databases.

C. Metabolomics (The Snapshot of State)

- Definition: The study of small molecules (metabolites) created during metabolism.
- Forensic Application: Determining the Post-Mortem Interval (PMI). By analyzing the "thanatometabolome" (metabolites produced after death), scientists can estimate time since death more accurately than via core temperature alone.
- Data Type: Concentration profiles, metabolic pathway maps.

D. Clinical Data

- Definition: Information collected during medical history, toxicology reports, and physical examinations.
- Forensic Application: Correlating biological omics data with toxicology (e.g., presence of drugs/poisons) to determine cause of death or impairment at the time of a crime.

2. Importance of Data in Modern Biology and Forensics

The shift from "observation-based" to "data-driven" forensics provides several advantages:

1. **Objectivity:** Quantitative data reduces the "examiner bias" often criticized in qualitative forensic disciplines (like pattern evidence).
2. **Predictive Power:** Using **Forensic DNA Phenotyping (FDP)**, we can provide leads on a suspect's appearance when there is no match in a database.
3. **Individualization:** Integrating multiple data types (e.g., matching a protein profile from a hair to a genomic profile from blood) increases the statistical certainty of an identity match.

3. Challenges in Biological Data Analysis

A. Data Fragmentation and Quality

- Degradation: Forensic samples are rarely pristine. Environmental factors (UV, heat, moisture) fragment DNA and denature proteins.

- Contamination: Mixture interpretation (data from multiple contributors) remains a complex computational challenge.

B. Storage and Processing (The Big Data Problem)

- A single High-Throughput Sequencing (HTS) run can generate terabytes of data. Forensic labs often lack the server infrastructure and bioinformaticians required to process this volume.

C. Ethical and Legal Hurdles

- Privacy: Genomic data can reveal predispositions to diseases or non-disclosed familial relationships (the "Golden State Killer" effect).
- Admissibility: New "Omics" techniques must meet the Daubert Standard (peer review, error rates, and general acceptance) before being used in court.

4. Opportunities: The Future of Forensic Science

1. The Molecular Autopsy: Using omics to identify "sudden unexplained deaths" (e.g., undiagnosed cardiac channelopathies) that traditional autopsies miss.
2. Microbiome Analysis: Using the "microbial fingerprint" of a person or a location to link a suspect to a crime scene (the Forensic Microbiome).
3. AI and Machine Learning: Developing algorithms that can scan massive multi-omic datasets to identify patterns invisible to the human eye, such as specific biomarkers for stress or physical exertion at the time of death.

Notes:

WHAT IS OMICS

Omics is a collective term for a broad field of biology that focuses on the comprehensive study of large sets of biological molecules that make up the structure, function, and dynamics of an organism.

The suffix "-omics" comes from the word genome (the complete set of genes) and implies the study of the entire collection of a specific type of molecule, rather than studying them individually.

Core Idea

Instead of looking at just one gene, protein, or metabolite at a time, "omics" technologies analyze ALL of them simultaneously to get a holistic, systems-level view of an organism's biology.

Think of it as:

Biology 1.0: Studying one tree.

Omics Biology: Using satellites and drones to map and analyze the entire forest, its ecosystem, and all its interactions.

Major Types of Omics Fields

Here are the most common and foundational ones:

Genomics: The study of the entire genome (all DNA, including genes and non-coding sequences).

Example: Sequencing a person's whole genome to find genetic variants linked to disease.

Transcriptomics: The study of the complete set of RNA transcripts (the transcriptome) produced by the genome under specific conditions.

Example: Comparing RNA from a cancer cell vs. a healthy cell to see which genes are "turned on" (expressed).

Proteomics: The study of the full set of proteins (the proteome) expressed by a cell, tissue, or organism. Proteins are the functional molecules that carry out the cell's work.

Example: Identifying all the proteins in a blood sample to find biomarkers for Alzheimer's disease.

Metabolomics: The study of all small-molecule metabolites (the metabolome), like sugars, lipids, and amino acids. These are the end-products of cellular processes and give a snapshot of an organism's physiology.

Example: Analyzing urine metabolites to understand how a person's body responds to a specific drug.

Epigenomics: The study of the complete set of epigenetic modifications (chemical tags on DNA or histones) that regulate gene expression without changing the DNA sequence itself.

Example: Mapping DNA methylation patterns across the genome to understand aging or environmental impacts.

Key Technologies

Omics relies heavily on high-throughput technologies:

Next-Generation Sequencing (NGS): For genomics and transcriptomics.

Mass Spectrometry (MS): For proteomics and metabolomics.

Bioinformatics & Big Data Analysis: Critical for storing, processing, and interpreting the massive, complex datasets ("Big Data") these techniques generate.

Applications of Omics

Precision Medicine: Tailoring medical treatment to an individual's unique genomic, proteomic, or metabolic profile.

Drug Discovery: Identifying new drug targets and understanding drug mechanisms.

Agriculture: Developing crops with better yield, nutrition, or stress resistance.

Microbiology: Studying microbial communities (microbiomes) in the gut, soil, or ocean.

Systems Biology: Integrating multiple omics datasets to create computer models of entire biological systems.

"Multi-Omics" Integration

The most powerful modern approach is multi-omics or integrative omics, which combines data from genomics, transcriptomics, proteomics, etc., to get a complete picture of the flow of biological information:

DNA (Genomics) → RNA (Transcriptomics) → Proteins (Proteomics) → Metabolites (Metabolomics)

This integration allows scientists to understand not just the parts list, but how all the parts work together in a complex system.

In short: Omics is the large-scale, data-driven study of all the molecules that define life, transforming biology and medicine from a science of individual parts to a science of complex, interconnected systems.

Discussion Questions for Students

- *How does the "Right to Privacy" conflict with the use of familial searching in genomic databases?*
- *Why might a protein profile be more useful than a DNA profile in a cold case involving 50-year-old skeletal remains?*